

Regression

Raffaele Saggio

UBC

August 26, 2026

Your life in the next 3-4 years

```
. reg price mpg, r
```

Linear regression

```
Number of obs   =      74
F(1, 72)         =     17.28
Prob > F         =     0.0001
R-squared        =     0.2196
Root MSE        =    2623.7
```

price	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
mpg	-238.8943	57.47701	-4.16	0.000	-353.4727	-124.316
_cons	11253.06	1376.393	8.18	0.000	8509.272	13996.85

Data is chaotic, need to make sense of this chaos



Regression must become your security blanket



Regression as Reduction

In order to arrive at a distinct formulation of statistical problems, it is necessary to define the task which the statistician sets himself: briefly, and in its most concrete form, the object of statistical methods is the reduction of data. A quantity of data, which usually by its mere bulk is incapable of entering the mind, is to be replaced by relatively few quantities which shall adequately represent the whole, or which, in other words, shall contain as much as possible, ideally the whole, of the relevant information contained in the original data. – R.A Fisher (1922)

Linear model

- Typical justification for OLS is a linear structural equation relating a dependent variable (the regressand) Y_i to a vector of independent variables (regressors) X_i :

$$Y_i = X_i' \beta + u_i$$

- Suppose $E[X_i u_i] = 0$.
- Then $E[X_i Y_i] = E[X_i X_i'] \beta$, so

$$\beta = E[X_i X_i']^{-1} E[X_i Y_i].$$

- This is the population OLS regression coefficient vector.
- Analogy principle $\implies$ estimate β by replacing population moments with sample moments, so $\hat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i Y_i \right)$.

Linear model

- Typical justification for OLS is a linear structural equation relating a dependent variable (the regressand) Y_i to a vector of independent variables (regressors) X_i :

$$Y_i = X_i' \beta + u_i$$

- Suppose $E[X_i u_i] = 0$.
- Then $E[X_i Y_i] = E[X_i X_i'] \beta$, so

$$\beta = E[X_i X_i']^{-1} E[X_i Y_i].$$

- This is the population OLS regression coefficient vector.
- Analogy principle $\implies$ estimate β by replacing population moments with sample moments, so $\hat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i Y_i \right)$.

Pop quiz: what do we mean by “linear structural equation relating [...]”?

$$Y_i = X_i' \beta + u_i$$

- We have seen in the previous class when a regression coefficient has a causal interpretation thanks to random assignment.
- Without random assignment, regression coefficient may or may not have a causal interpretation as we shall see.
- Setting causality aside, regression estimates have “mechanical” statistical properties that are important to know that are intimately related with a fundamental statistical object: the CEF.

- Recall the **conditional expectation** of Y_i given X_i :

$$E[Y_i | X_i = x] = \int_{-\infty}^{\infty} y dF_{Y|X}(y | X_i = x)$$

- The conditional expectation function (CEF), $\mu(x) = E[Y_i | X_i = x]$, describes how the conditional mean of Y_i changes as we vary x
- The CEF can be defined for any Y_i and X_i . It has no behavioral/causal content per se.
- Yet, it is often the starting point to establish the existence of “statistical relationships” between variables.

Summarizing a complex world...

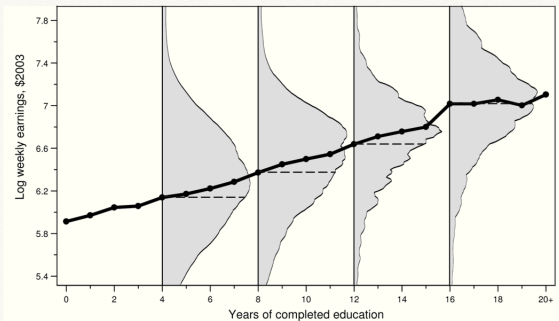


Figure 3.1.1: Raw data and the CEF of average log weekly wages given schooling. The sample includes white men aged 40-49 in the 1980 IPUMS 5 percent file.

CEF Residuals

- We can always write

$$Y_i = \mu(X_i) + u_i$$

- Here $u_i \equiv Y_i - \mu(X_i)$.
- By taking conditional expectations on both sides we obtain

$$E[Y_i|X_i] = \mu(X_i) + E[u_i|X_i]$$

- Since $E[Y_i|X_i] = \mu(X_i)$, we have $E[u_i|X_i] = 0$.
- Note that the CEF residual (or “error term”) u_i has conditional mean zero by definition.
- We have also that
 - 1 $E[u_i] = 0$.
 - 2 $E[h(X_i)u_i] = 0$.

Optimal Predictor

- Suppose we want to use X_i to predict Y_i
- Formally, consider the following minimization problem:

$$\min_{m \in \mathcal{M}} E \left[(Y_i - m(X_i))^2 \right]$$

- Here $\mathcal{M}$ is the set of all measurable functions mapping $\mathcal{R}^K$ to $\mathcal{R}^1$
- This is a complicated problem since we are optimizing over all possible functions

Optimal Predictor

- Rewrite the problem

$$\begin{aligned} & \min_{m \in \mathcal{M}} E \left[([Y_i - \mu(X_i)] + [\mu(X_i) - m(X_i)])^2 \right] \\ &= \min_{m \in \mathcal{M}} E \left[(Y_i - \mu(X_i))^2 \right] + 2E[(Y_i - \mu(X_i))(\mu(X_i) - m(X_i))] \\ & \quad + E[(\mu(X_i) - m(X_i))^2] \end{aligned}$$

- First term doesn't depend on $m(\cdot)$ and the second is zero (why?), so this is equivalent to solving

$$\min_{m \in \mathcal{M}} E[(\mu(X_i) - m(X_i))^2]$$

CEF as Optimal Predictor

$$\min_{m \in \mathcal{M}} E \left[(\mu(X_i) - m(X_i))^2 \right]$$

- This is minimized at zero by setting $\mu(X_i) = m(X_i)$.
- In other words, the CEF is the minimum mean squared error (MMSE) predictor of Y_i .

ANOVA - Law of Total Variance

- Law of total variance

$$\text{Var}(Y_i) = \underbrace{\text{Var}[E[Y_i|X_i]]}_{\text{Variability in explained components}} + \underbrace{E[\text{Var}[Y_i|X_i]]}_{\text{Variability in Unexplained Components}}$$

- Important to study inequality (wage inequality, health inequality, etc...).
- Well defined statistical decomposition.

ANOVA - Law of Total Variance

- Law of total variance

$$\text{Var}(Y_i) = \underbrace{\text{Var}[E[Y_i|X_i]]}_{\text{Variability in explained components}} + \underbrace{E[\text{Var}[Y_i|X_i]]}_{\text{Variability in Unexplained Components}}$$

- Important to study inequality (wage inequality, health inequality, etc...).
- Well defined statistical decomposition.
- Ok, great but who tells me $\mu(X_i)$?

Linear Projections

- Suppose we want to predict Y_i , but limit ourselves to linear functions
- The **linear projection** of Y_i on X_i is $X_i'\beta$, where

$$\beta \equiv \arg \min_b E \left[(Y_i - X_i'b)^2 \right]$$

- First-order conditions: $E [X_i (Y_i - X_i'\beta)] = 0$
- Then

$$\beta = E [X_i X_i']^{-1} E [X_i Y_i]$$

- This is the OLS coefficient vector
- The OLS projection $X_i'\beta$ is the MMSE **linear** predictor of Y_i .
- We define $E^*[Y|X] = X_i'\beta$.

Linear Projections and the CEF

- By applying the same trick we saw with the CEF, we can rewrite

$$\begin{aligned}\beta &\equiv \arg \min_b E \left[(Y_i - X_i' b)^2 \right] \\ &= \arg \min_b E \left[(\mu(X_i) - X_i' b)^2 \right]\end{aligned}$$

- OLS therefore also gives the MMSE linear prediction of the **CEF** of Y_i given X_i
- Note: This also implies that when the CEF is linear, OLS fits it exactly

BLP $\rightarrow$ Linear approximation of the CEF

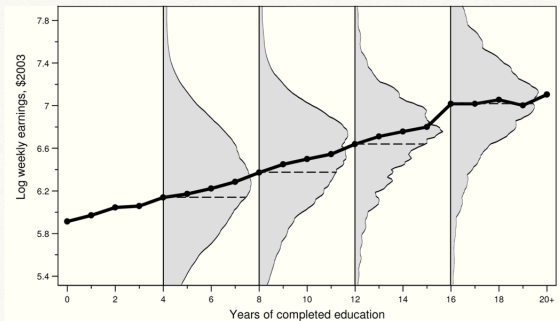


Figure 3.1.1: Raw data and the CEF of average log weekly wages given schooling. The sample includes white men aged 40-49 in the 1980 IPUMS 5 percent file.

BLP $\rightarrow$ Linear approximation of the CEF

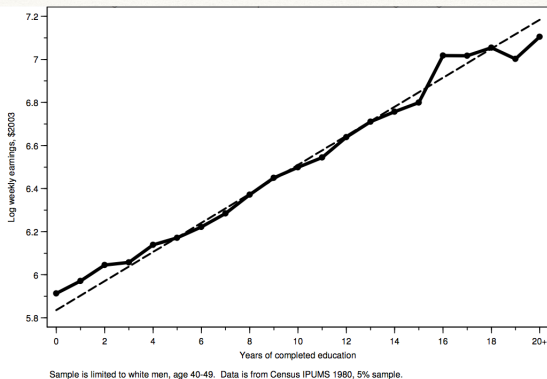


Figure 3.1.2: Regression threads the CEF of average weekly wages given schooling

- Projection model:

$$Y_i = X_i' \beta + U_i$$

- Here $U_i \equiv Y_i - X_i' \beta$ is defined as the difference between Y_i and the least squares projection
- Note that

$$\begin{aligned} E[X_i U_i] &= E[X_i (Y_i - X_i' \beta)] \\ &= 0. \end{aligned}$$

- Note: the projection errors U_i are defined so that X_i and U_i are uncorrelated. They have no “life of their own”.

Omitted Variable Bias

- Consider two regression models:

$$Y_i = \alpha_s + \beta_s S_i + u_{is}$$

$$Y_i = \alpha_\ell + \beta_\ell S_i + \gamma_\ell A_i + u_{i\ell}$$

- Example: “Short” regression of log earnings on schooling, vs. “long” regression of log earnings on schooling and ability (SAT score)

Omitted Variable Bias

$$Y_i = \alpha_s + \beta_s S_i + u_{is}$$

$$Y_i = \alpha_\ell + \beta_\ell S_i + \gamma_\ell A_i + u_{i\ell}$$

- We might prefer the long regression coefficient β_ℓ
- Perhaps higher-ability people go to school longer, but schooling is as good as randomly assigned among people with the same ability
- In other words, the CEF of Y_i given (S_i, A_i) might be more economically interesting than the CEF of Y_i given S_i

Omitted Variables Bias

- Short regression coefficient:

$$\begin{aligned}\beta_s &= \frac{\text{Cov}(Y_i, S_i)}{\text{Var}(S_i)} \\ &= \frac{\text{Cov}(\alpha_\ell + \beta_\ell S_i + \gamma_\ell A_i + u_{i\ell}, S_i)}{\text{Var}(S_i)} \\ &= \beta_\ell + \gamma_\ell \frac{\text{Cov}(A_i, S_i)}{\text{Var}(S_i)}\end{aligned}$$

- The short coefficient equals the long coefficient, plus the coefficient on the omitted variable times the regression of the omitted variable on the included variable.
- This is the standard “omitted variables bias” (OVB) formula. It shows how OLS coefficients change when more regressors are added.
- The long regression coefficient is not necessarily better. Again, it depends on which CEF is of interest.

Good, Bad and Ugly Controls



Good or Bad?

- You want to study whether women are systematically discriminated in the labor market.
- Regression of earnings on gender using age and race as controls returns a coefficient of -10,000 dollars.
- But then you run the same regression also controlling for occupation. Coefficient on gender dummy shrinks to zero.
- Is occupation a good control variable in this context?

Good, Bad and Ugly Controls

- Good controls:
 - Within each value of control, the treatment is as good as randomly assigned.
 - Fixed at the time in which treatment was assigned.
- Bad controls: more is not always better!
 - Variables that should be considered as outcomes per se!
 - Contaminates interpretation of the treatment.
- Ugly controls: what most of us ended up dealing with
 - Need to argue why they are good controls in observational data.
 - Lots of them can create statistical/inference problems → what variation drives my “treatment” (next classes + FWL)?

- Let's now think about how to obtain an estimate of β from a single sample.

$$Y_i = X_i' \beta + u_i \quad (1)$$

- Analogy principle!

$$\hat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i Y_i \right) \quad (2)$$

- Let's now think about how to obtain an estimate of β from a single sample.

$$Y_i = X_i' \beta + u_i \quad (1)$$

- Analogy principle!

$$\hat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i Y_i \right) \quad (2)$$

- We will now think about some key algebraic properties of the LS solution (finite sample perspective).

Projection Matrices

- Define the projection matrix:

$$P_X = X(X'X)^{-1}X'$$

- The OLS predicted values are $\hat{y} = X\hat{\beta} = P_X y$.
- P_X projects any vector into the column space of X .
- Likewise, $M_X = I - P_X$ projects any vector into the orthogonal complement. The OLS residual is $\hat{u} = M_X y$.
- P_X is called the “hat” matrix since it “puts a hat” on y , while M_X is the “residual maker” since it turns y into a residual orthogonal to the columns of X

Projection Matrices are awesome! Pt.1

- The diagonal element of the matrix P_X is sometimes called the statistical leverage, call the i th element P_{ii} .
- Important property: $P_{ii} \in [0; 1]$.
- The statistical leverage measures how influential a given observation i in deriving $\hat{\beta}$.

$$P_{ii} = \frac{\partial \hat{y}_i}{\partial y_i} \quad (3)$$

- “Balanced design” means that all observations have equal P_{ii} .
- How influential is observation i for my OLS coefficients?

$$\begin{aligned} \hat{\beta}_{-i} &= (\sum X_{-i} X'_{-i})^{-1} (\sum X_{-i} Y_{-i}) \\ &= \hat{\beta} - (\sum X_i X'_i)^{-1} \left(\sum X_i \frac{\hat{u}_i}{(1 - P_{ii})} \right) \end{aligned} \quad (4)$$

$P_{ij} = 1 \rightarrow$ Overfitting



Projection Matrices are awesome Pt.2

- Useful properties of projection matrices:
 - Symmetric: $P_X = P_X'$, $M_X = M_X'$
 - Idempotent: $P_X P_X = P_X$, $M_X M_X = M_X$
 - Zero product: $P_X M_X = 0$
 - $P_X X = X$, and $M_X X = 0$ (M_X “annihilates” X)

Frisch-Waugh-Lovell Theorem

- Consider the regression model:

$$y = X_1\beta_1 + X_2\beta_2 + u$$

- We can write $y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{u}$, where $\hat{\beta}_1$ and $\hat{\beta}_2$ are OLS estimates and $\hat{u}$ is the residual
- Instead of running this regression, let's first take residuals from regressions of y and X_2 on X_1 :

$$\tilde{y} = M_1y, \tilde{X}_2 = M_1X_2$$

- Then regress $\tilde{y}$ on $\tilde{X}_2$:

$$\tilde{y} = \tilde{X}_2\gamma_2 + u_2$$

Frisch-Waugh-Lovell Theorem

- OLS estimate of γ_2 :

$$\begin{aligned}\hat{\gamma}_2 &= (\tilde{X}_2' \tilde{X}_2)^{-1} \tilde{X}_2' \tilde{y} \\ &= (X_2' M_1 M_1 X_2)^{-1} X_2' M_1 M_1 y \\ &= (X_2' M_1 X_2)^{-1} X_2' M_1 (X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + \hat{u}) \\ &= (X_2' M_1 X_2)^{-1} (X_2' M_1 X_1 \hat{\beta}_1 + X_2' M_1 X_2 \hat{\beta}_2 + X_2' M_1 \hat{u})\end{aligned}$$

Frisch-Waugh-Lovell Theorem

$$\hat{\gamma}_2 = (X_2' M_1 X_2)^{-1} (X_2' M_1 X_1 \hat{\beta}_1 + X_2' M_1 X_2 \hat{\beta}_2 + X_2' M_1 \hat{u})$$

- The first term in parentheses is zero
- $M_1 \hat{u} = \hat{u}$ and $X_2' \hat{u} = 0$ so the third term is also zero. Then

$$\begin{aligned}\hat{\gamma}_2 &= (X_2' M_1 X_2)^{-1} X_2' M_1 X_2 \hat{\beta}_2 \\ &= \hat{\beta}_2.\end{aligned}$$

- The “long regression” coefficient on X_2 , $\hat{\beta}_2$, is therefore the same as the “residual regression” coefficient, $\hat{\gamma}_2$. This is the first part of the FWL theorem.

Frisch-Waugh-Lovell Theorem

- The second part of the FWL theorem is obtained by multiplying both sides of our expression for y by M_1 :

$$\begin{aligned}M_1 y &= M_1 X_1 \hat{\beta}_1 + M_1 X_2 \hat{\beta}_2 + M_1 \hat{u} \\ \implies \tilde{y} &= \tilde{X}_2 \hat{\beta}_2 + M_1 \hat{u}\end{aligned}$$

- Since $\hat{\beta}_2$ is the coefficient from the residual regression, $M_1 \hat{u}$ is the residual from this regression. Since $M_1 \hat{u} = \hat{u}$, the residual regression generates the same residuals as the long regression. This is the second part of the FWL theorem.
- This result means that standard errors obtained from the residual regression will also match the long regression (with an appropriate degrees of freedom correction)

Frisch-Waugh-Lovell Theorem: Summary

- Together, the two parts of the FWL theorem say that if we're interested in a subset of the coefficients from a multivariate regression, we can first “partial out” the other variables from both sides, then work with the residuals.
- This procedure will generate both the same coefficients and the same residuals we'd get from the long regression.
- Other notes:
 - We can also obtain the same coefficients by first partialling out some variables from the right-hand side but not the left. This will not produce the same residuals, however.
 - In general, we will NOT obtain the same coefficients by partialling out variables from the left but not the right.

Analysis of Variance

- Now that we understand the mechanics of regression, let's consider a few ways it's used in practice
- One traditional use, proposed by R.A. Fisher, is “Analysis of Variance” (ANOVA)
- ANOVA simply asks whether the mean of some variable differs across observed groups
- Name derives from fact that in linear model can partition variance of response into components due to “fixed” predictors $X'\beta$ and error.

Analysis of Variance

- Suppose we have data on wages w_i for a sample of individuals
- We also have data on race ($black_i \in \{0, 1\}$) and gender ($female_i \in \{0, 1\}$)
- Suppose we run the regression

$$w_i = \alpha_0 + \alpha_1 black_i + \alpha_2 female_i + \alpha_3 black_i \cdot female_i + u_i$$

- How should we interpret the coefficients $(\alpha_0, \alpha_1, \alpha_2, \alpha_3)$?

Analysis of Variance

$$w_i = \alpha_0 + \alpha_1 \text{black}_i + \alpha_2 \text{female}_i + \alpha_3 \text{black}_i \cdot \text{female}_i + u_i$$

- This is a **saturated** model: The number of coefficients equals the number of possible values of the right-hand-side variables
- In a saturated model the CEF is necessarily linear
- As we've learned, regression perfectly fits the CEF when the CEF is linear

Analysis of Variance

$$w_i = \alpha_0 + \alpha_1 \text{black}_i + \alpha_2 \text{female}_i + \alpha_3 \text{black}_i \cdot \text{female}_i + u_i$$

- Then

$$\alpha_0 = E[w_i | \text{black}_i = 0, \text{female}_i = 0]$$

$$\alpha_1 = E[w_i | \text{black}_i = 1, \text{female}_i = 0] - E[w_i | \text{black}_i = 0, \text{female}_i = 0]$$

$$\alpha_2 = E[w_i | \text{black}_i = 0, \text{female}_i = 1] - E[w_i | \text{black}_i = 0, \text{female}_i = 0]$$

$$\alpha_3 = (E[w_i | \text{black}_i = 1, \text{female}_i = 1] - E[w_i | \text{black}_i = 1, \text{female}_i = 0]) \\ - (E[w_i | \text{black}_i = 0, \text{female}_i = 1] - E[w_i | \text{black}_i = 0, \text{female}_i = 0])$$

- Any question about differences in means across groups can be answered with these parameters

Getting Practical

- We are interested in studying the relationship between Y and X . X is (basically) continuous, thus it is hard to estimate CEF.
- Example: Y_c share of uninsured people living in area c . X_c is average income in area c .
- Can run a regression but we are worried that the relationship might not be linear (why?).
- What would you do?

classes/aux/scat1.jpg

classes/aux/scat2.jpg

classes/aux/scat3.jpg

classes/aux/scat4.jpg

classes/aux/scat5.jpg

classes/aux/scat6.jpg

What if we want to plot this relationship after having controlled for some variables?

What if we want to plot this relationship after having controlled for some variables?

classes/aux/cattaneo_max.jpg

Getting Practical

- Lots of textbooks (1st grad econometrics) might have told you that running a regression is fine only if the outcome is “continuous” (e.g. log wage).

Getting Practical

- Lots of textbooks (1st grad econometrics) might have told you that running a regression is fine only if the outcome is “continuous” (e.g. log wage).
- This recommendation is misguided and you shall not follow it.
- If the interest lies in assessing the (average) causal impact of some treatment D on outcome Y , running a regression of Y on D will return the ATE when there is random assignment of D , regardless of whether Y is continuous, discrete, binary.
- The same does not hold if running a probit regression on Y on D (and you have random assignment of D)!.

Getting Practical

- Often you encounter an outcome variable where for lots of observations are such that $Y_i = 0$.

Getting Practical

- Often you encounter an outcome variable where for lots of observations are such that $Y_i = 0$.
- Provided that $Y_i > 0$, Y_i then looks like a continuous variable.
 - 1 Provide me an example of such variable.
 - 2 A lot of people in these situations run a regression of $\log(Y_i)$ on D_i . Is that regression identifying the causal effect of D_i on Y_i ?
 - 3 Other people instead run a regression of $\log(1 + Y_i)$ on D_i ? What do you think is the rationale behind that?

classes/aux/zeros.jpg

Connecting Regression with Treatment Effects

- Suppose we run randomized experiments at several sites
- Treatment $T_i \in \{0, 1\}$ is randomly assigned within a site, but the probability of assignment differs across sites
- We therefore only want to compare treatment and control groups within sites
- Let $X_i \in \{1 \dots K\}$ denote site, and let $D_i = (1 \{X_i = 1\}, \dots, 1 \{X_i = K\})'$ denote a vector of site indicators

Stratified Experiments

- We run the OLS regression:

$$Y_i = D_i' \Pi + \delta T_i + u_i$$

- This model includes a dummy for every site, but only a main effect of the treatment $\rightarrow$ Is this regression saturated?
- What does δ identify in terms of potential outcomes (Y_{i1}, Y_{i0}) ?

Stratified Experiments

- We run the OLS regression:

$$Y_i = D_i' \Pi + \delta T_i + u_i$$

- This model includes a dummy for every site, but only a main effect of the treatment $\rightarrow$ Is this regression saturated?
- What does δ identify in terms of potential outcomes (Y_{i1}, Y_{i0}) ?
- Recall that we can always write

$$Y_i = D_i Y_{i1} + (1 - D_i) Y_{i0} \tag{5}$$

Stratified Experiments

$$Y_i = D_i' \Pi + \delta T_i + u_i$$

- Let's use our knowledge of the FWL theorem to derive an expression for δ :

$$\delta = E [\tilde{T}_i^2]^{-1} E [\tilde{T}_i \tilde{Y}_i]$$

- $\tilde{T}_i$ and $\tilde{Y}_i$ are residuals from regressions on D_i
- We can rewrite δ as follows... why?

$$\delta = \frac{E [(Y_i - E[Y_i|X_i]) (T_i - E[T_i|X_i])]}{E [(T_i - E[T_i|X_i])^2]}$$

Stratified Experiments

$$\delta = \frac{E[Y_i(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]} - \frac{E[E[Y_i|X_i](T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]}$$

- The second term is zero (why?)
- By iterated expectations the first term can be written

$$\delta = \frac{E[E[Y_i|X_i, T_i](T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]}$$

Stratified Experiments

- It's useful to rewrite

$$E[Y_i|X_i, T_i] = E[Y_i|X_i, T_i = 0] + \Delta(X_i)T_i,$$

$$\Delta(X_i) = E[Y_i|X_i, T_i = 1] - E[Y_i|X_i, T_i = 0]$$

- Here $\Delta(X_i)$ is a site-specific treatment/control contrast.
- We can then write (why?):

$$E[Y_i|X_i, T_i = 0] = E[Y_{i0}|X_i] \tag{6}$$

$$\Delta(X_i) = E[Y_{i1} - Y_{i0}|X_i] \tag{7}$$

- Putting pieces together now...

$$\begin{aligned}
 \delta &= \frac{E[(E[Y_i|X_i, T_i = 0] + \Delta(X_i)T_i)(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]} \\
 &= \frac{E[\Delta(X_i)T_i(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]} \\
 &= \frac{E[E[Y_{i1} - Y_{i0}|X_i]T_i(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]}
 \end{aligned}$$

Stratified Experiments

- By invoking iterated expectations one more time we obtain

$$\delta = \frac{E[E[Y_{i1} - Y_{i0}|X_i]E[T_i(T_i - E[T_i|X_i])|X_i]]}{E[E[(T_i - E[T_i|X_i])^2|X_i]]}$$

Stratified Experiments

- By invoking iterated expectations one more time we obtain

$$\begin{aligned}\delta &= \frac{E[E[Y_{i1} - Y_{i0}|X_i]E[T_i(T_i - E[T_i|X_i])|X_i]]}{E[E[(T_i - E[T_i|X_i])^2|X_i]]} \\&= \frac{E[E[Y_{i1} - Y_{i0}|X_i]\sigma_T^2(X_i)]}{E[\sigma_T^2(X_i)]} \\&= E[E[Y_{i1} - Y_{i0}|X_i]w(X_i)]\end{aligned}$$

- Here $w(X_i) = \sigma_T^2(X_i) / E[\sigma_T^2(X_i)]$
- Since treatment is binary $\sigma_T^2(X_i) = \text{Pr}[T_i = 1|X_i](1 - \text{Pr}[T_i = 1|X_i])$
- OLS with controls for site indicators therefore identifies a variance-of-treatment-weighted average of site-specific treatment/control comparisons
- Works even in the context of misspecification of the CEF!
- Useful statistical object even when T_i is not randomly assigned within groups

Going one step beyond

- New scenario: two treatments are now being randomized within the site.
- Run the following regression

$$Y_i = D_i\Pi + T_{i,1}\delta_1 + T_{i,2}\delta_2 + u_i \quad (8)$$

- Randomization implies $\rightarrow (Y_{i0}, Y_{i1}) \perp (T_{i,1}, T_{i,2}) | D_i$
- What is (δ_1, δ_2) identifying?

Contamination Bias in Linear Regressions*

Paul Goldsmith-Pinkham
Yale University

Peter Hull
Brown University

Michal Kolesár
Princeton University

June 2, 2022

Abstract

We study the interpretation of regressions with multiple treatments and flexible controls. Such regressions are often used to analyze stratified randomized control trials with multiple intervention arms, to estimate value-added (for, e.g., teachers) with observational data, and to leverage the quasi-random assignment of decision-makers (e.g. bail judges). We show that these regressions generally fail to estimate convex averages of heterogeneous treatment effects, even when the treatments are conditionally randomly assigned and the controls are sufficiently flexible to avoid omitted variables bias. Instead, estimates of each treatment's effects are generally contaminated by a non-convex average of the effects of other treatments. Thus, recent concerns about heterogeneity-induced bias in regressions leveraging potential outcome restrictions (e.g. parallel trends assumptions) also arise with “design-based” identification strategies. We discuss solutions to the contamination bias and propose a new class of efficient estimators of weighted average effects that avoid bias. In a re-analysis of the Project STAR trial, we find minimal bias because treatment effect heterogeneity is largely idiosyncratic. But sizeable contamination bias arises when effect heterogeneity becomes correlated with treatment propensity scores.

Where is the failure?

$$\delta_1 = E [\tilde{T}_i^2]^{-1} E [\tilde{T}_i \tilde{Y}_i]$$

- where now $\tilde{T}_i$ are residuals from regressions of T_i on D_i and $T_{i,2}$. Similarly for $\tilde{Y}_i$.
- Observed outcome now can be written as

$$Y_i = Y_{i0} + \underbrace{(Y_{i1} - Y_{i0})}_{\tau_{i1}} T_{i,1} + \underbrace{(Y_{i2} - Y_{i0})}_{\tau_{i2}} T_{i,2} \quad (9)$$

- Same tricks as before and we arrive at

$$\begin{aligned} \delta_1 &= \frac{E [Y_i (T_{i1} - E^* [T_{i1} | X_i, T_{i2}])]}{E [(T_{i1} - E^* [T_{i1} | X_i, T_{i2}])^2]} \\ &= \frac{E [Y_{i0} (T_{i1} - E^* [T_{i1} | X_i, T_{i2}])]}{E [(T_{i1} - E^* [T_{i1} | X_i, T_{i2}])^2]} \\ &\quad + \frac{E [\tau_{i1} T_{i,1} (T_{i1} - E^* [T_{i1} | X_i, T_{i2}])]}{E [(T_{i1} - E^* [T_{i1} | X_i, T_{i2}])^2]} \\ &\quad + \frac{E [\tau_{i2} T_{i,2} (T_{i1} - E^* [T_{i1} | X_i, T_{i2}])]}{E [(T_{i1} - E^* [T_{i1} | X_i, T_{i2}])^2]} \end{aligned}$$

- First term $\rightarrow$
- Second term $\rightarrow$
- Third term $\rightarrow$

Table 1: Project STAR Contamination Bias and Treatment Effect Estimates

	A. Contamination Bias Estimates				
	Regression	Own	Bias	Worst-Case Bias	
	Coefficient	Effect		Negative	Positive
	(1)	(2)	(3)	(4)	(5)
Small Class Size	5.357 (0.778)	5.202 (0.778)	0.155 (0.160)	-1.654 (0.185)	1.670 (0.187)
Teaching Aide	0.177 (0.720)	0.360 (0.714)	-0.183 (0.149)	-1.529 (0.176)	1.530 (0.177)
	B. Treatment Effect Estimates				
	Unweighted	Efficiently-Weighted			
	(ATE)	One-at-a-time	Common		
	(1)	(2)	(3)		
Small Class Size	5.561 (0.763) [0.744]	5.295 (0.775) [0.743]	5.563 (0.764) [0.742]		
Teaching Aide	0.070 (0.708) [0.694]	0.263 (0.715) [0.691]	-0.003 (0.712) [0.695]		

Notes: Panel A estimates the contamination bias and range of potential contamination bias in regression estimates of small class and teaching aide treatment effects for the Project STAR kindergarten analysis. The analysis sample includes 5,868 students. Column 1 reports estimates from a partially linear model in eq. (25). Columns 2 and 3 estimate the own- and cross-treatment decomposition of this estimate in eq. (27). Columns 4 and 5 reports the smallest (largest) possible contamination bias from reordering the conditional ATEs to be as negatively (positively) correlated with the cross-treatment weights as possible. Panel B summarizes estimates of small class and teaching aide treatment effects from different specifications in the kindergarten sample of Project STAR. Column 1 reports estimates of small class and teaching aide treatment effects from the interacted model in eq. (28). Column 2 reports estimates from the treatment-specific regressions in eq. (29). Column 3 reports estimates from the efficiently weighted specification, using the estimated weights in eq. (30). Robust standard errors are reported in parentheses. Standard errors that assume the propensity scores are known are reported in square brackets.